Ashish Kumar Dash

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Skills

Languages: C++, JavaScript, Java, PHP
Frameworks: React, Node.js, Next.js, Tailwinds
Tools/Platforms: MongoDB, MySQL, Mathematics, Maven, OS, Computer Networks
Soft Skills: Quick-Learning, Adaptability, Problem-Solving, Decision-Making

Projects

Trie Top-K Autocomplete System | Mar' 2026

- Developed an interactive Autocomplete System using Trie data structure to efficiently retrieve and display Top-K suggestions based on user input prefixes for fast and accurate search experiences.
- Introduced advanced features including frequency-based ranking, LRU caching, and dynamic updates. Utilised STL data structures for efficient prefix traversal, cache management, and real-time query optimisation.
- Constructed a performance analysis module to evaluate key metrics like query latency, cache hit rate, and memory usage, enabling direct comparison of system efficiency under varying workloads.
  Tech Used: C++, STL, Trie Data Structure, LRU Cache, Multithreading, File Handling.

2D Game Engine with Classic Game Implementations | Feb' 2026

- Developed a modular game engine-like system in C++ to support multiple classic games such as Pong, Tetris, Snake, Connect Four, and Conway's Game of Life, focusing on reusable low-level components and efficient system design.
- Implemented core engine modules including game loop, rendering pipeline, input handling system, collision detection, and state management, utilizing object-oriented design principles for extensibility and separation of concerns.
- Constructed a scalable architecture enabling multiple games to run on shared infrastructure, optimizing memory usage, frame updates, and real-time performance across different gameplay simulations.
  Tech Used: C++, STL, Object-Oriented Programming (LLD principles), raylib (rendering and input handling), Algorithms, File Handling.

CPU-Scheduling | Dec' 2025

- Developed an interactive CPU Scheduling Simulator to visually demonstrate and compare complex Operating System process allocation strategies for educational and analytical purposes.
- Introduced diverse scheduling algorithms, including Round Robin, FCFS, SJN, and SRTF. Used Java Collections for efficient management of process queues and execution simulation logic.
- Constructed a performance analysis module to automatically calculate and compare key metrics like turnaround time, waiting time, and CPU utilisation, enabling the direct evaluation of algorithmic efficiency through Gantt charts.
  Tech Used: Java, Java Swing and AWT, Java Collections.

Training

Summer Training | W3Grads | MERN Stack & AI Integration Trainee | Jun' 2025 – Aug' 2025

- Completed an intensive training program focused on the MERN stack, mastering the integration of Artificial Intelligence APIs and models into modern web architectures.
- Made a scalable, full-stack web application from scratch, implementing responsive front-end designs and robust back-end logic infused with AI-driven features like a chatbot.

Executed the full software development lifecycle (SDLC), demonstrating proficiency in version control

Certificates

- MERN Stack Development | W3Grads | Aug' 2025 | Aug' 2025
- Introduction to Hardware and Operating Systems | Coursera | Sep' 2024
- The Bits and Bytes of Computer Networking | Coursera | Sep' 2024

Achievements

Solved 500+ Coding Problems: Mar' 2026

Education

Lovely Professional University Jalandhar, Punjab
Bachelor of Technology Aug' 2023 - Present
Computer Science and Engineering**: CGPA: 8.4**

Christ H.S.S. Cuttack, Odisha
Intermediate May' 2022 – May' 2023

S.S.V.M Cuttack, Odisha
Matriculation May' 2020 – May'2021